

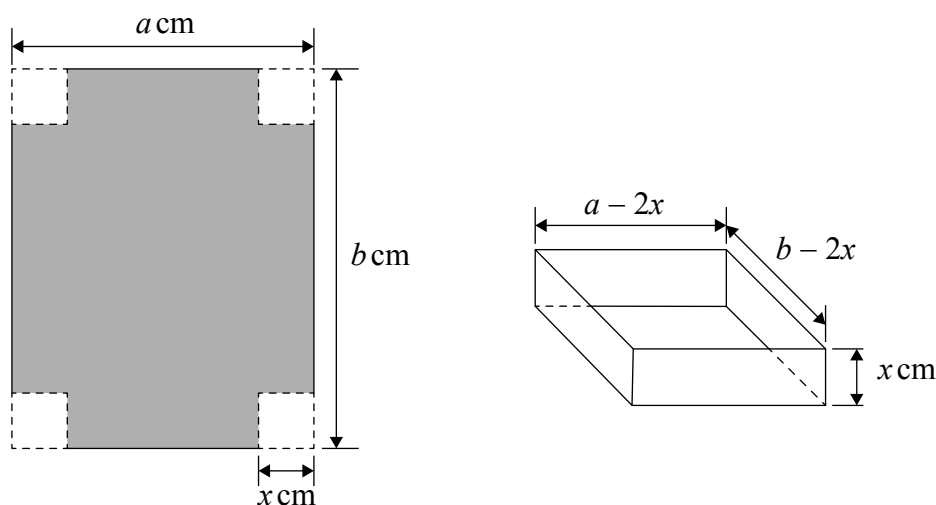
2. [Maximum mark: 28]

This question asks you to consider open-topped boxes formed by cutting square corners from an $a \times b$ sheet of cardboard and folding the sides. In particular, you are asked to investigate conditions for which the maximum volume of a box is an integer.

A rectangular sheet of cardboard measures a cm by b cm where $a < b$.

A square of side length x cm is cut from each corner and the resulting sheet of cardboard is then folded to form an open-topped box with a rectangular base.

This is shown in the following diagrams.



The volume, $V \text{ cm}^3$, of a box is given by $V = x(a - 2x)(b - 2x)$ where $0 \leq x \leq \frac{a}{2}$.

(a) Write down the value of V when

(i) $x = 0$; [1]

(ii) $x = \frac{a}{2}$. [1]

(b) Show that $V = 4x^3 - 2(a + b)x^2 + abx$. [2]

(c) Show that the only solutions of $\frac{dV}{dx} = 0$ are $x = \frac{(a + b) \pm \sqrt{a^2 - ab + b^2}}{6}$. [6]

(This question continues on the following page)

(Question 2 continued)

In parts (d) and (e), the following three results are given.

$$x = \frac{(a+b) + \sqrt{a^2 - ab + b^2}}{6} \text{ lies outside the interval } 0 \leq x \leq \frac{a}{2}.$$

$$x = \frac{(a+b) - \sqrt{a^2 - ab + b^2}}{6} \text{ lies inside the interval } 0 \leq x \leq \frac{a}{2}.$$

$$\sqrt{a^2 - ab + b^2} > 0.$$

Let $x_m = \frac{(a+b) - \sqrt{a^2 - ab + b^2}}{6}.$

(d) Use $\frac{d^2V}{dx^2}$ to show that there is a local maximum of V at $x = x_m$. [5]

(e) Use your answers to parts (a) and (d) to explain why x_m will give the maximum volume, V_m , of the box. [1]

In parts (f) to (h), consider a method for generating sets of positive integer values for x_m and hence V_m .

Consider the case where $a = 2st - s^2$, $b = t^2 - s^2$, and $s, t \in \mathbb{Z}^+$.

(f) Given that $a < b$, show that $t > 2s$. [2]

It is given that $\sqrt{a^2 - ab + b^2} = s^2 - st + t^2$. You are **not** required to show this result.

(g) By substituting for each of a , b and $\sqrt{a^2 - ab + b^2}$ in terms of s and t , show that $x_m = \frac{s(t-s)}{2}$. [3]

(h) (i) Given that $a, b \in \mathbb{Z}^+$, use $V_m = x_m(a - 2x_m)(b - 2x_m)$ to explain why s being even implies that V_m is an integer. [2]

(ii) State another condition on s and a corresponding condition on t , that implies that both x_m and V_m are integers. [1]

(iii) Hence, or otherwise, find an integer value of x_m and the corresponding integer value of V_m . [4]